

**Virginia Electric and Power Company
Surry Power Station
5570 Hog Island Road
Surry, Virginia 23883**

November 11, 2021

U. S. Nuclear Regulatory Commission
Attention: Document Control Desk
Washington, D. C. 20555-0001

Serial No.: 21-343
SPS: MMT
Docket No.: 50-280
50-281
License No.: DPR-32
DPR-37

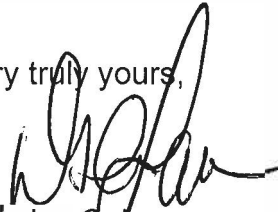
Dear Sir or Madam:

Pursuant to 10CFR50.73, Virginia Electric and Power Company hereby submits the following Licensee Event Report applicable to Surry Power Station Unit 1 and Unit 2

Report No. 50-280 / 2021-001-00

This report has been reviewed by the Station Facility Safety Review Committee and will be forwarded to the Management Safety Review Committee.

Very truly yours,



Douglas Q. Lawrence
Site Vice President
Surry Power Station

Enclosure

Commitment contained in the LER: None

cc: U.S. Nuclear Regulatory Commission, Region II
Marquis One Tower, Suite 1200
245 Peachtree Center Ave., NE
Atlanta, GA 30303-1257

NRC Senior Resident Inspector
Surry Power Station



LICENSEE EVENT REPORT (LER)

(See Page 3 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk all: oir_submission@omb.eop.gov. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. Facility Name Surry Power Station, Unit 1	2. Docket Number 05000 280	3. Page 1 OF 3
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4. Title
Unanalyzed Condition Due to Appendix R Concern Identified With Cable Separation

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name	Docket Number
09	13	2021	2021	- 001 -	00	11	11	2021	Surry Power Station, Unit 2	05000 281
									Facility Name	Docket Number
										05000

9. Operating Mode N	10. Power Level 100
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11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)

☐ 10 CFR Part 20	☐ 20.2203(a)(2)(vi)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	10 CFR Part 73
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(4)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.71(a)(5)
☐ 20.2203(a)(2)(i)	10 CFR Part 21	☐ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(1)(i)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(i)
☐ 20.2203(a)(2)(iii)	10 CFR Part 50	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.77(a)(2)(ii)
☐ 20.2203(a)(2)(iv)	☐ 50.36(c)(1)(i)(A)	☒ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	
☐ 20.2203(a)(2)(v)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	
☐ OTHER (Specify here, in abstract, or NRC 366A).				

12. Licensee Contact for this LER

Licensee Contact Douglas C. Lawrence, Site Vice President	Phone Number (Include area code) (757) 365-2001
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13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS

14. Supplemental Report Expected

☒ No	☐ Yes (If yes, complete 15. Expected Submission Date)
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15. Expected Submission Date

Month	Day	Year

16. Abstract (Limit to 1560 spaces, i.e., approximately 15 single-spaced typewritten lines)

On September 13, 2021, at 1822 EDT, an apparent non-compliance with 10 CFR 50, Appendix R, Section III.G.2 (separation of redundant fire safe shutdown equipment) was identified based on Operating Experience (OE) from North Anna Power Station (LER 2021-002-00). A review of the affected control circuits for the Unit 1 and Unit 2 Emergency Diesel Generator (EDG) output breakers and Emergency Bus feeder breakers identified a concern that damage to the control power circuits during a fire may affect the ability to provide emergency power on both units. An 8-hour non-emergency report was made per 10 CFR 50.72(b)(3)(ii)(B) for an unanalyzed condition. Fire watches were established in the affected areas. The cause of this condition is the station's original post-fire safe shutdown analysis did not accurately assess the impact on the equipment from associated cable damage due to fire for the select cables. Risk mitigating actions have been developed, implemented, and will continue to be performed until appropriate permanent corrective actions are in place to mitigate this condition. This event is reportable in accordance with 10 CFR 50.73(a)(2)(ii)(B) for "Any event or condition that resulted in the nuclear power plant being in an unanalyzed condition that significantly degraded plant safety."



LICENSEE EVENT REPORT (LER) CONTINUATION SHEET

(See NUREG-1022, R.3 for instruction and guidance for completing this form
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1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
Surry Power Station, Unit 1	05000-280	YEAR 2021	SEQUENTIAL NUMBER 001	REV NO. 00

NARRATIVE

Plant Operating Conditions Prior to the Event

Unit 1 - 100 Percent Power
Unit 2 - 100 Percent Power

1.0 Description of Event

On September 13, 2021, at 1822 EDT, an apparent non-compliance with 10 CFR 50, Appendix R, Section III.G.2 (separation of redundant fire safe shutdown equipment) was identified based on Operating Experience (OE) from North Anna Power Station (LER 2021-002-00). A review of the affected control circuits for the Unit 1 and Unit 2 Emergency Diesel Generator (EDG) output breakers (EIS Component BKR, System EK) and Emergency Bus feeder breakers (EIS Component BKR, System EK) identified a concern that damage to the control power circuits during a fire may affect the ability to provide emergency power on both units. An 8-hour non-emergency report was made per 10 CFR 50.72(b)(3)(ii)(B) for an unanalyzed condition (EN #55459). This event is reportable in accordance with 10 CFR 50.73(a)(2)(ii)(B) for "Any event or condition that resulted in the nuclear power plant being in an unanalyzed condition that significantly degraded plant safety."

Several control cables were identified within common fire areas that, if fire damaged, could prevent the availability of both trains of Emergency Diesel Generators (EDGs) (EIS Component DG, System EK) to start and power their respective emergency buses (EIS Component BU, System EB) required by the station's current Appendix R analysis. The following issues were found:

1. Fire damage to control power cables for normal supply breakers may result in loss of control power and inability to power the emergency bus without manual operator actions outside of the main control room.
2. Hot shorts on the normal supply breakers (EIS Component BKR, System EK) could result in spurious breaker operations that may prevent EDGs and/or emergency buses from performing its Appendix R function.

2.0 Significant Safety Consequences and Implications

No significant safety consequences resulted from this event. Surry Appendix R analysis assumes all equipment within the fire area or zone is damaged; however, it is expected that damage would be minimized due to the defense in depth strategy for fire protection. Installed fire protection features are provided within the areas of concern to ensure fire damage to the impacted control cabling is minimized. All affected fire areas have installed area-wide fire detection and installed suppression systems. During this timeframe, Surry Units 1 and 2 operated at 100% power. The health and safety of the public were not affected by this event.

3.0 Cause of the Event

The cause of this condition is the station's original post-fire safe shutdown analysis did not accurately assess the impact on the equipment from associated cable damage due to fire for the select cables. In addition, the original Multiple Spurious Operations review effort failed to recognize applicability of EDG/Emergency Power scenarios for some areas for the plant.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

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Surry Power Station, Unit 1		05000-	280	YEAR	SEQUENTIAL NUMBER	REV NO.
				2021	001	00

NARRATIVE**4.0 Immediate Corrective Action**

Fire watches were established in the Unit 1 Normal Switchgear Room, Unit 1 Cable Tray Room, Unit 1 Emergency Switchgear Room, and Unit 2 Cable Vault and Tunnel to mitigate the risk of fire and aid in early detection of fire. Risk mitigating actions have been developed, implemented, and will continue to be performed until appropriate permanent corrective actions are in place to mitigate this condition.

5.0 Additional Corrective Actions

A Reasonable Assurance of Safety (RAS) was developed with risk mitigating actions implemented until final resolution is completed. Procedures ECA-0.0 (Loss of All AC Power) and Flex Support Guidelines (FSGs) were reviewed to mitigate the results of a loss of all AC power conditions. Periodic reviews of applicable procedures and Fire Fighting Strategies for fires in areas of concern have been established. Prior to any unavailability of the Turbine Driven Auxiliary Feedwater (TDAFW) pump (E1IS Component P, System BA), guidance has been provided to pre-stage the Beyond Design Basis (BDB) Auxiliary Feedwater Pump.

The station is currently evaluating design changes to eliminate the vulnerability.

6.0 Actions to Prevent Recurrence

The applicable control cables associated with the EDG output breakers and Emergency Bus normal feeder breakers are planned to be modified or manual actions are planned to be established, where applicable, to regain compliance as required by Appendix R Safe Shutdown Analysis.

7.0 Similar Events

A review of internal operating experience was performed, and no similar events were identified that were previously reported to the Nuclear Regulatory Commission.

8.0 Additional Information

Units 1 and 2 continued to operate at 100% power.